SAE./No.232/April 2023

Studies in Applied Economics

UNITED KINGDOM: A GOLDEN GROWTH RATE AND CREDIT COUNTERPARTS ANALYSIS

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Johns Hopkins Institute for Applied Economics,
Global Health, and the Study of Business Enterprise

United Kingdom: A Golden Growth Rate and Credit Counterparts Analysis

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The *Studies in Applied Economics* series is under the general direction of Prof. Steve H. Hanke, Founder and Co-Director of the Johns Hopkins Institute for Applied Economics, Global Health and the Study of Business Enterprise (hanke@jhu.edu). The views expressed in each working paper are those of the authors and not necessarily those of the institutions' that the authors are affiliated with.

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Abstract

This paper conducts a Golden Growth Rate (GGR) analysis for the United Kingdom (UK) for the 2010-2019 period. A credit counterpart analysis for the 2010-2022 period is also presented to examine the monetary growth of the Bank of England and its monetary policy.

SECTION 1: Golden Growth Analysis[1]

The GGR is the growth rate of the money supply that would allow the central bank to hit its inflation target. The UK's GGR covers the 2010-2019 period and disregards the subsequent years to avoid the effects of the central bank's stimulus program caused by the COVID-19 pandemic.

To calculate the GGR, we used the income form of the Quantity Theory of Money (QTM). The QTM states that MV = Py, where "M" is the money supply, "V" is the velocity of money, "P" is the price level, and "y" is real GDP.

The percent-change form of the QTM is used to calculate the GGR: $\Delta \boldsymbol{M} + \Delta \boldsymbol{V} = \Delta \boldsymbol{P} + \Delta \boldsymbol{y}$,[2] or, equivalently, $\Delta \boldsymbol{M} = \Delta \boldsymbol{P} + \Delta \boldsymbol{y} - \Delta \boldsymbol{V}$. $\Delta \boldsymbol{M}$ is the estimated GGR of the broad money supply, M4x,[3] to keep inflation at the target rate. $\Delta \boldsymbol{P}$ is the central bank's inflation target.[4] $\Delta \boldsymbol{y}$ is the average of the YoY change in real GDP over the period.[5]

To calculate $\Delta \boldsymbol{V}$, first divide each year's nominal GDP[6] by its broad money supply and obtain the money velocity for the 2009-2019 period as V = $\frac{Year-end\ Nominal\ GDP}{Year-end\ M4x}$. Then, calculate the YoY change of money velocity for the 2010-2019 period. Lastly, an average of money velocity throughout the duration results in the change of money velocity, $\Delta \boldsymbol{V}$.

SECTION 2: UK's Golden Growth Rate Calculation

The Bank of England's target inflation is 2% and used as $\Delta \boldsymbol{P}$. From 2010 through 2019, the UK's average annual real GDP growth was 2.03%, and the average change of money velocity was -0.04%. Thus, based on the percent-change-form of the QTM, the GGR for M4x is $\Delta \boldsymbol{M} = \Delta \boldsymbol{P} + \Delta \boldsymbol{y} - \Delta \boldsymbol{V}$ = 2% + 2.03% - (-0.04%) = 4.07% per year. The actual M4x growth during the same duration was 3.81%, implying that **the Bank of England has been increasing its money supply nearly in par with its golden growth rate**. Not surprisingly, the UK's average actual annual inflation between 2010-2019 (2.01%) was almost the same as the Bank of England's target of 2%.

The graphs below compare the UK's golden growth rate with its actual annual money supply (M4x) growth (Figure 1), and its inflation target with its realized inflation rate (Figure 2).

[1] For treatment of the Quantity Theory of Money as it relates to Golden Growth Rates, see Greenwood-Hanke (2021) https://onlinelibrary.wiley.com/doi/abs/10.1111/jacf.12479

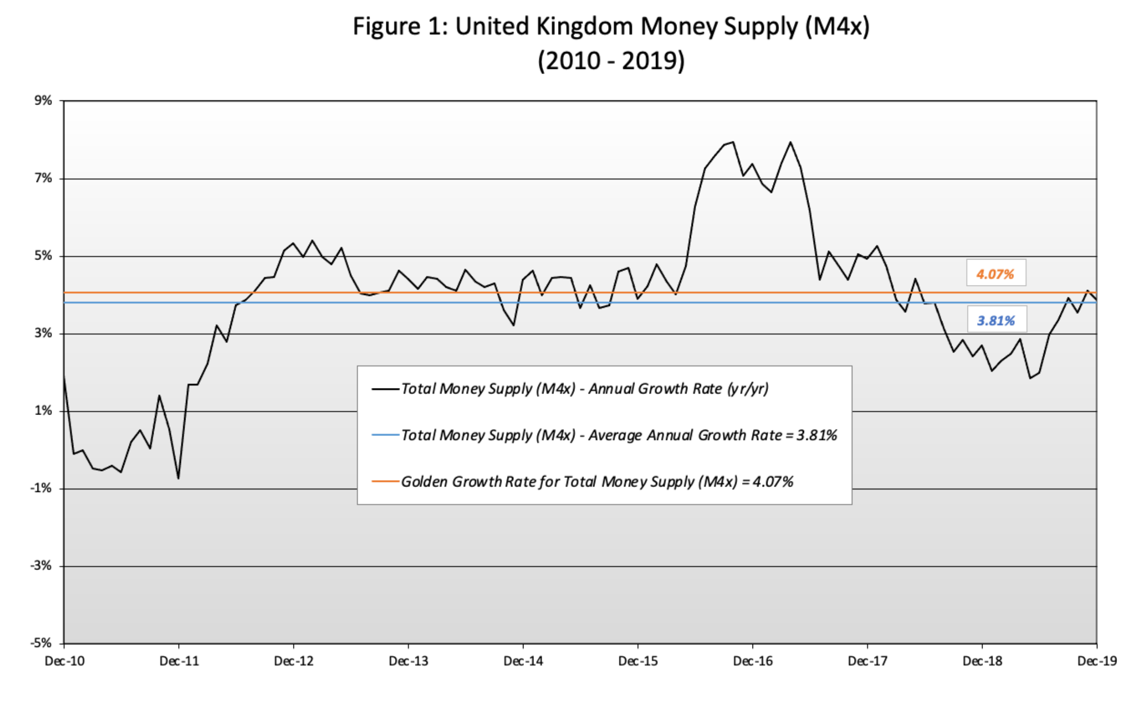
[2] For a derivation of the percent change form, see: Friedman 1987 https://miltonfriedman.hoover.org/internal/media/dispatcher/214346/full

[3] Bank of England. Further details about M4 excluding intermediate other financial corporations (OFCs) data. https://www.bankofengland.co.uk/statistics/details/further-details-about-m4-excluding-intermediate-other-financial-corporations-data

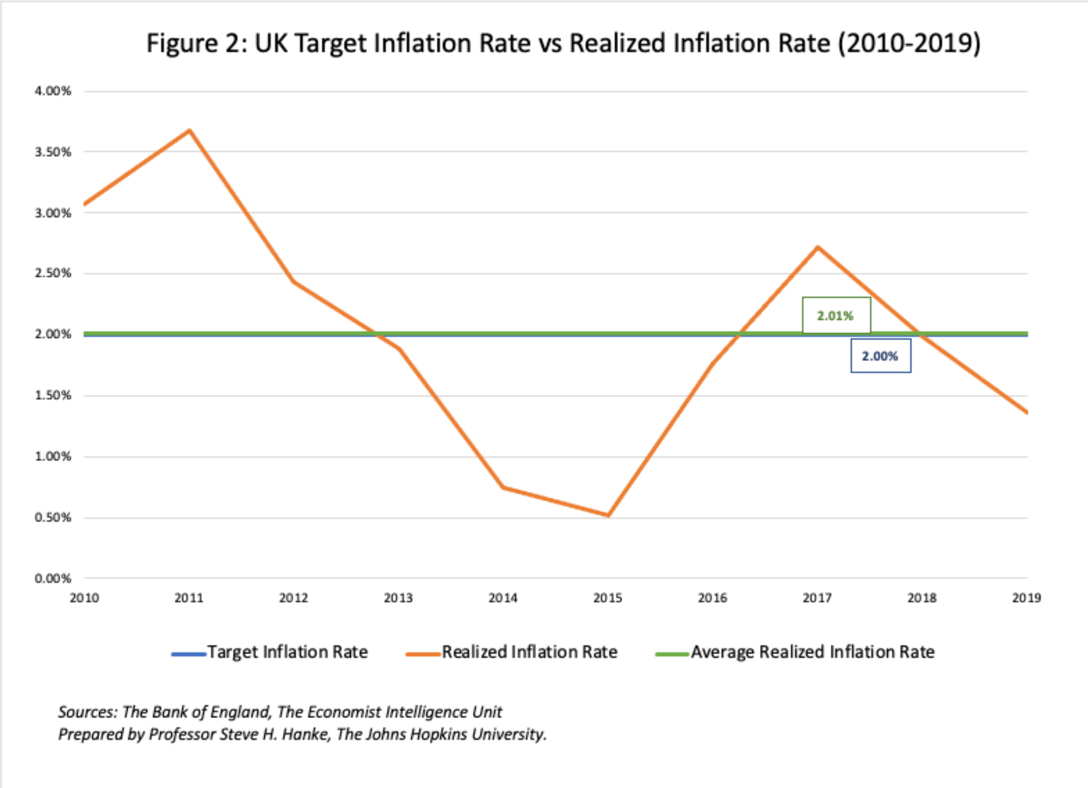
[4] Bank of England. Inflation and the 2% target. https://www.bankofengland.co.uk/monetary-policy/inflation

[5] Economist Intelligence Unit. Real GDP (% change pa). https://viewpoint.eiu.com/data/results?searchId=889550fb-938c-4367-9f76-9fd25669d3e1

[6] Economist Intelligence Unit. Nominal GDP (LCU). https://viewpoint.eiu.com/data/results?searchId=2d588e88-aa34-469a-bf4e-581d6cdc72ca



Sources: The Bank of England, Refinitiv Datastream, The Economist Intelligence Unit.
Prepared by Prof. Steve H. Hanke, The Johns Hopkins University.



SECTION 3: Credit Counterparts Analysis[7]

Credit counterpart analysis examines the factors that drive the money supply change. The broad money supply (here, M4x) is composed of currencies issued by the Bank of England and deposits from the commercial banks.[8] While currencies are a liability of the central bank, they are also an asset of the nonbank public by the credit counterpart. Similarly, deposits are liabilities to commercial banks, but assets to the nonbank public. Thus, counterparts of these liabilities that show up on the central bank's balance sheet equal the sum of commercial bank lending, plus the reserves of banks held at the central bank, plus residual items that fill the discrepancy. In terms of changes, both the liability side of the central bank and asset side of commercial banks can be expressed as the following:

Liability side (of the central bank):

1. Δ Broad Money = Δ Cash in circulation of the public + Δ Bank deposits

Asset side (of commercial banks):

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[7] For logic of the Credit Counterpart Analysis, see: Hanke (2017). https://www.e-elgar.com/shop/usd/money-in-the-great-recession-9781784717827.html

[8] The Bank of England defines M4x as the sum of: (1) households' M4 (seasonally adjusted) + (2) private non-financial corporations' M4 (seasonally adjusted) + (3) non-intermediate other financial corporations' M4 (seasonally adjusted). https://www.bankofengland.co.uk/statistics/details/further-details-about-m4-excluding-intermediate-other-financial-corporations-data

2. Δ Broad Money = Δ Commercial Bank Lending + Δ Commercial Bank reserves at the central bank +/- Δ Other Items (net)

A credit counterpart analysis determines the change in money supply (M4x) from the asset side of commercial banks. In this memo, both annual and monthly data from 2010 through 2022 are presented. The annual data outline an overview of the Bank of England's monetary operations, while the monthly data provide a more detailed look into changes that enable us to analyze the impact of significant events such as the European Debt Crisis, Brexit Referendum, and the COVID-19 pandemic.

The assets of the UK's commercial banks are categorized as follows:

(1) **Reserves (commercial bank deposits at the central bank)**
The Bank of England uses the term "Reserve balances" that are "*weekly amounts outstanding of Central Bank sterling reserve balance liabilities*." This can be found from the Bank of England's *Bankstats tables* at the B1.1.2 section with the code B56A.[9]

(2) **Lending (Loans)**
The Bank of England refers to commercial bank lending as "*Monthly amounts outstanding of UK resident monetary financial institutions' (excluding the Central Bank) sterling net lending to the private sector excluding intermediate OFCs*." This can be found from the Bank of England's *Bankstats tables* at the A2.2.3 section with the code B57Q.[10]

(3) **Residuals**
This is the difference between the money supply (M4x) and the previous two categories (Reserves and Lending). The money supply (M4x) is referred to as "*Monthly amounts outstanding of M4x (monetary financial institutions' sterling M4 liabilities to private sector excluding intermediate OFCs)*." This can be found from the Bank of England's *Bankstats tables* at the A2.2.3 section with the code B53Q.[11]

Figure 3 below shows the UK's M4x absolute levels and its three components that together contribute to M4x growth from 2010 through 2022. Figure 4 then shows the annual growth of M4x and its components in percentage form and how each component contributes to M4x. For instance, in 2020, M4x increased by 13.84%. The contributors to this change were: bank lending

[9] Bank of England Database. Weekly amounts outstanding of Central Bank sterling reserve balance liabilities (in sterling millions).
https://www.bankofengland.co.uk/boeapps/database/fromshowcolumns.asp?Travel=NIxSUx&FromSeries=1&ToSeries=50&DAT=RNG&FD=1&FM=Jan&FY=2010&TD=31&TM=Dec&TY=2022&FNY=&CSVF=TT&html.x=89&html.y=26&C=IRU&Filter=N

[10] Bank of England Database. Monthly amounts outstanding of UK resident monetary financial institutions' (excl. Central Bank) sterling net lending to Private sector excluding intermediate OFCs (in sterling millions).
https://www.bankofengland.co.uk/boeapps/database/fromshowcolumns.asp?Travel=NIxSUx&FromSeries=1&ToSeries=50&DAT=RNG&FD=1&FM=Jan&FY=2010&TD=31&TM=Dec&TY=2022&FNY=&CSVF=TT&html.x=150&html.y=5&C=JVI&Filter=N

[11] Bank of England Database. Monthly amounts outstanding of UK resident monetary financial institutions' sterling M4 liabilities to Private Sector excluding intermediate OFCs (in sterling millions).
https://www.bankofengland.co.uk/boeapps/database/fromshowcolumns.asp?Travel=NIxSUx&FromSeries=1&ToSeries=50&DAT=RNG&FD=1&FM=Jan&FY=2010&TD=31&TM=Dec&TY=2022&FNY=&CSVF=TT&html.x=64&html.y=7&C=JVC&Filter=N

(3.89%), reserves (13.08%), and residuals (-3.12%) that together sum to the M4x growth rate (13.84%).

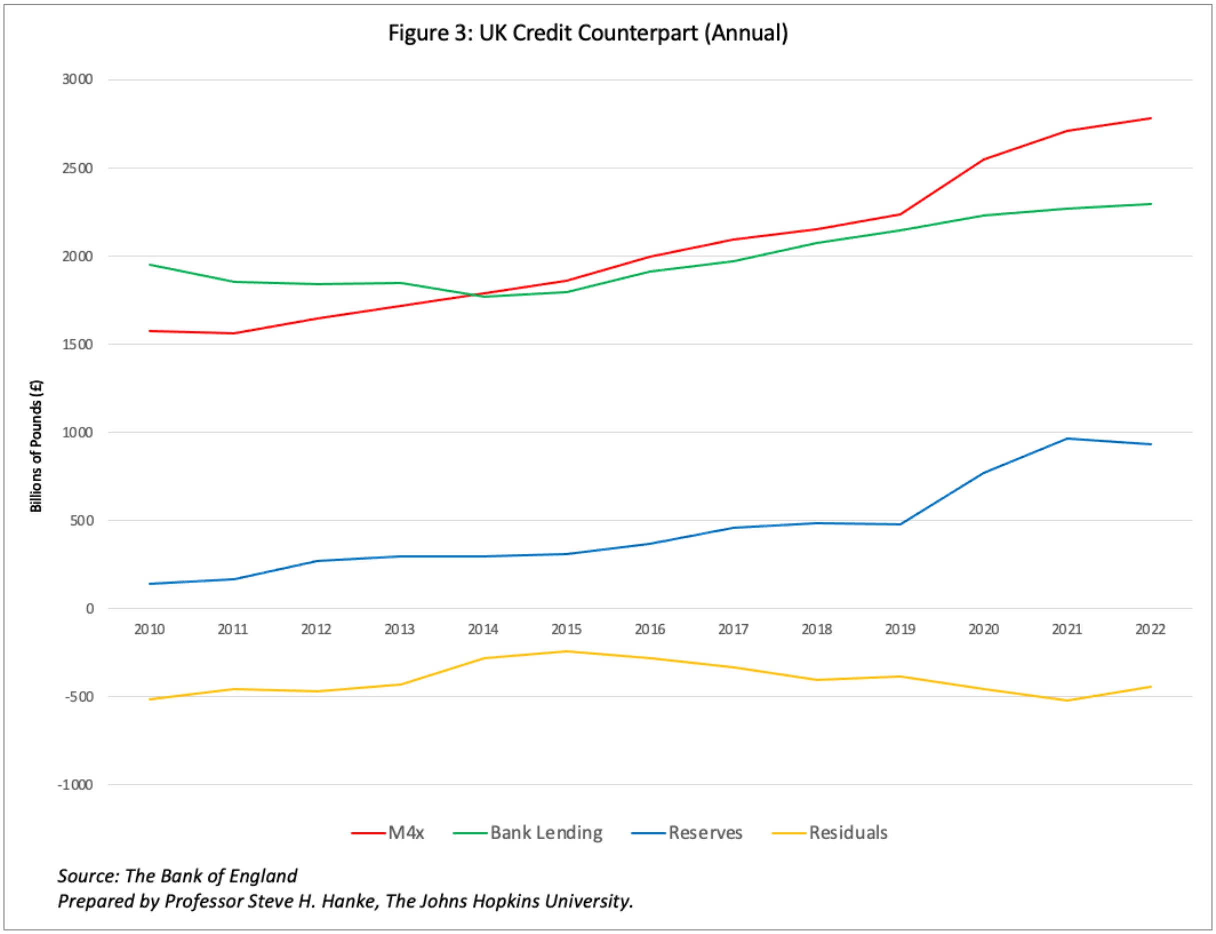
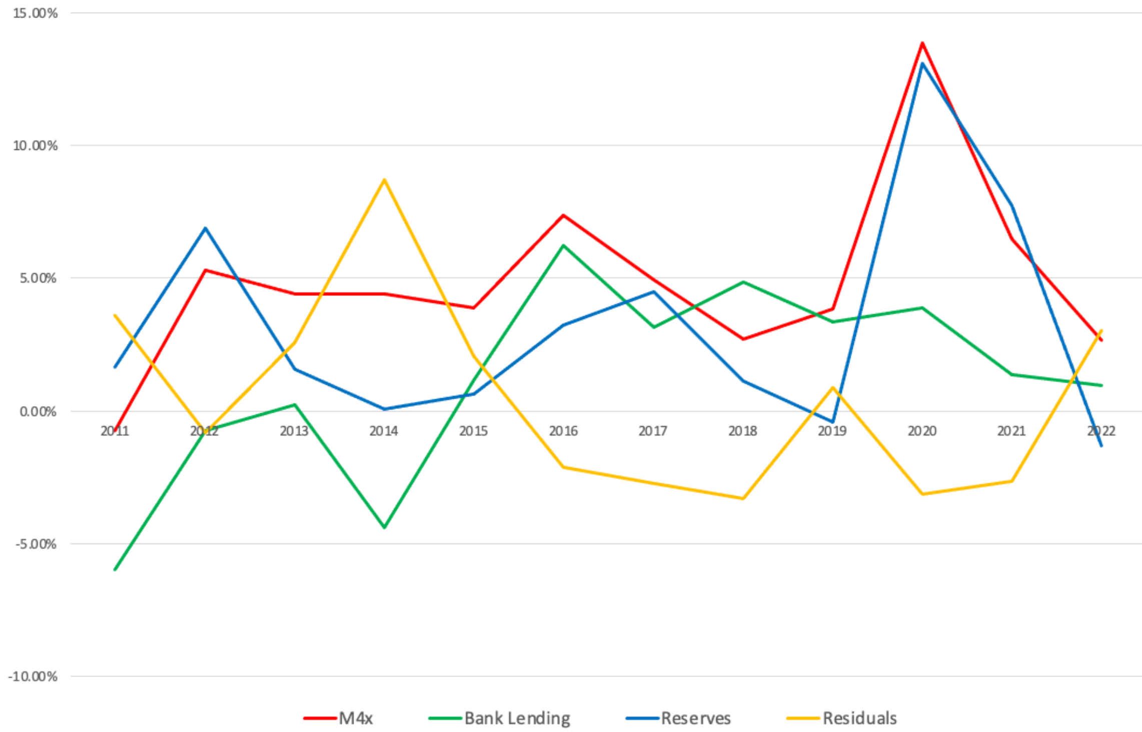


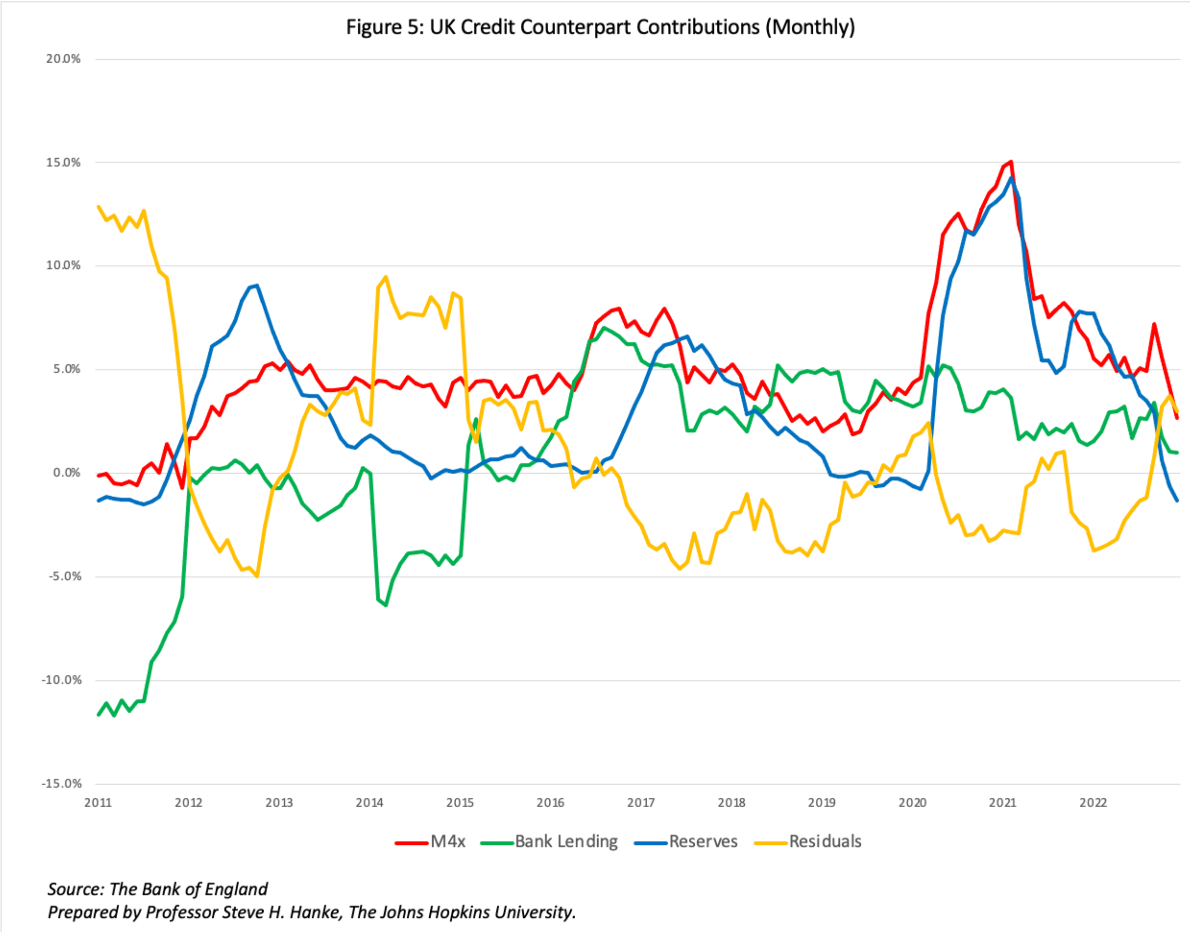
Figure 4: UK Credit Counterpart Contributions (Annual)



Source: The Bank of England
Prepared by Professor Steve H. Hanke, The Johns Hopkins University.

SECTION 4: The Bank of England's Monetary Operations

To better understand the Bank of England's monetary policies and the reasons behind the changes in its money supply, we have analyzed the Bank of England's monthly M4x change with its credit counterpart components as presented in Figure 5 below and provided background information with four key stages.



1. **European Debt Crisis QE2 and QE3 (Quantitative Easing)**, 2011-2012[12]

In response to the Euro Crisis, the Bank of England initiated QE2 in October 2011 and completed 125 billion pounds of gilts purchases in May 2012. With the economic outlook still gloomy and in an attempt to prevent a double-dip recession, the central bank decided to purchase an additional 50 billion pounds of gilts in July 2012 that lasted until November 2012. As Figure 5 shows, this quantitative easing led to a dramatic increase in the contribution of reserves to M4x in October 2011. The contribution reached its peak with a 9.1% contribution in October 2012 when it was the biggest contributor to increases in M4x. As the concerns about the European Debt Crisis faded, and the QE2 and QE3 periods came to an end, the monthly growth in reserves dropped to 0% at the end of 2014.

2. **Brexit Referendum Stimulus**, 2015-2017

To mitigate the market turmoil from the UK's vote to leave the European Union, the Bank of England announced that it would allow banks to reduce their amount of capital holdings and

[12] Bank of England. "The Bank of England's unconventional monetary policies: why, what and how". https://www.ecb.europa.eu/events/pdf/conferences/130607/PanelDiscussion_joyce.pdf?9b7c697afa3b9dce804543b2b4538d5b

lend as much as 150 billion pounds to households and corporations in July 2016.[13] Additionally, it deployed 60 billion pounds of government bond purchases over six months, starting in August 2016.[14] As Figure 5 indicates, in June 2016, the monthly change of M4x surpassed 6%, mainly driven by commercial bank lending.

3. **COVID-19 Pandemic QE**, 2020-2021[15]

To support and save the economy from the COVID-19 Pandemic, the Bank of England offered unlimited QE in March 2020. The amount of QE during the pandemic is estimated to be around 450 billion pounds. As demonstrated in Figure 5, the monthly change in reserves began to increase in April 2020 and skyrocketed to 14.3% in February 2021 with the growth of M4x reaching its peak around the same time.

4. **QT (Quantitative Tightening)**, 2021-2022[16]

As the inflation rate soared from the massive QE program, the Bank of England, for the first time since the COVID-19 pandemic outbreak, hiked its interest rate from its historic low of 0.1% in December 2021 to 0.25%. Additionally, the Bank of England begun to sell some of its bonds in November 2022 to unwind QE. While the Bank of England started the process of QT in February 2022 by not reinvesting maturing bonds, November's operation was the first "active" QT through bond selling. As presented in Figure 5, from January 2022 to December 2022, the rate of change in reserves continuously fell. Furthermore, reserves contracted at an unprecedented pace, hitting a first negative growth (-0.6%) in November 2022, driving a fall in M4x.

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[13] Chattanooga Times Free Press. "The Latest: UK central bank acts to support lending". https://www.timesfreepress.com/news/2016/jul/05/latest-uk-central-bank-acts-support-lending/

[14] Reuters. "Bank of England wields stimulus 'sledgehammer' to beat Brexit Blues". https://www.reuters.com/article/us-britain-eu-boe/bank-of-england-wields-stimulus-sledgehammer-to-beat-brexit-blues-idUSKCN10E2SX

[15] Bloomberg. "Bank of England Chief Economist Suggests Pandemic QE a Mistake". https://www.bloomberg.com/news/articles/2022-11-08/bank-of-england-chief-economist-suggests-pandemic-qe-a-mistake

[16] Financial Times. "Bank of England begins selling bonds as it unwinds QE Programme". https://www.ft.com/content/64e5dcdd-42a8-4013-a3bd-75a62cf8e6dc

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